Forecasting Volatility and Tail Risk During Extreme Market Events

Complete Research Paper Writing and Execution Plan

GARCH-EVT | Realized GARCH | Quantile Regression

Purpose of this document
This document converts the current project state into a publication-oriented execution plan. It separates what is already robust from what must still be corrected or re-estimated, freezes the research questions, defines the manuscript architecture, and sets an exact order for final model reruns, results freezing, writing, and reproducibility checks.

Prepared from the current Google Drive code, documentation, result tables, forecast files, and figure set.

Contents

1. Executive assessment of the project

The project has moved beyond a simple comparison of three models. It now contains a strong asymmetric daily-volatility benchmark, a conditional EVT extension, a realized-information volatility model, and two direct quantile-regression specifications, all evaluated across six major equity indices.

The empirical design is already much richer than the original outline: daily histories reach back to 1990, realized measures begin in the 2011-2013 period depending on market, the common post-2013 evaluation window contains multiple stress episodes, and the data pipeline explicitly protects the tail observations rather than cleaning them away.

Current project status

The project is best viewed as roughly 90% model-complete but not yet results-frozen. The remaining work is mainly publication-defensibility: equalize information sets, remove the remaining look-ahead channels in Realized GARCH preprocessing/estimation, strengthen inference, correct a small number of data and metadata issues, regenerate one authoritative result release, and only then draft the final Results and Abstract.

Operational model set

Family	Final specification	What it genuinely forecasts
Daily parametric benchmark	AR(1)-GJR-GARCH-skew-t	Conditional volatility + parametric VaR
Conditional EVT	GJR-GARCH + POT/GPD residual tail	Same volatility as GJR + EVT VaR/ES
Realized-information model	Log-linear Realized GARCH	Conditional volatility + parametric VaR/ES
Direct quantile model	QR-Full	Conditional VaR only
Sparse direct quantile model	QR-Range	Conditional VaR only

An important implementation detail is that quantile regression does not produce a genuine conditional variance forecast. Its SigmaHat field is reconstructed only to satisfy the common forecast-file schema, so QR must never be scored with QLIKE, RMSE, or other volatility-loss functions.

2. Refined title and conceptual framing

The current title should be broadened from 'forecasting volatility' to 'forecasting volatility and tail risk' because the model families do not all target the same object.

Recommended title

Forecasting Volatility and Tail Risk During Extreme Market Events: A Multi-Market Comparison of GARCH-EVT, Realized GARCH, and Quantile Regression

The title makes the two forecast targets explicit:

$$\text{Conditional variance:} \quad h_{(t+1)}$$

$$\text{Conditional downside tail risk:} \quad Q_{\alpha}(r_{(t+1)} \mid F_t)$$

This framing prevents a conceptual error that would otherwise run through the paper: QR estimates conditional quantiles directly, whereas GARCH-EVT and Realized GARCH route through a conditional variance.

3. The project is really two experiments

Experiment A: volatility forecasting

Primary question: Does high-frequency realized information improve next-day conditional-variance forecasts?

GJR-skew-t versus Realized GARCH

GARCH-EVT does not form a separate volatility model because it reuses the GJR conditional variance by construction. Quantile regression is excluded because it does not produce a genuine variance forecast.

Experiment B: tail-risk forecasting

Primary question: Which modelling philosophy best estimates extreme downside conditional quantiles?

GJR-skew-t | GARCH-EVT | Realized GARCH | QR-Full | QR-Range

Expected shortfall comparisons are narrower because only GARCH-EVT and Realized GARCH currently produce genuine ES forecasts.

Why the two-lane design matters
A model can be very good at forecasting conditional scale and still place the extreme quantile in the wrong location. Keeping the variance lane and the tail-risk lane separate is not cosmetic; it is the conceptual backbone of the paper.

4. The central scientific story

The strongest current empirical pattern is not that one model dominates every metric. It is that volatility accuracy and extreme-tail calibration rank models differently.

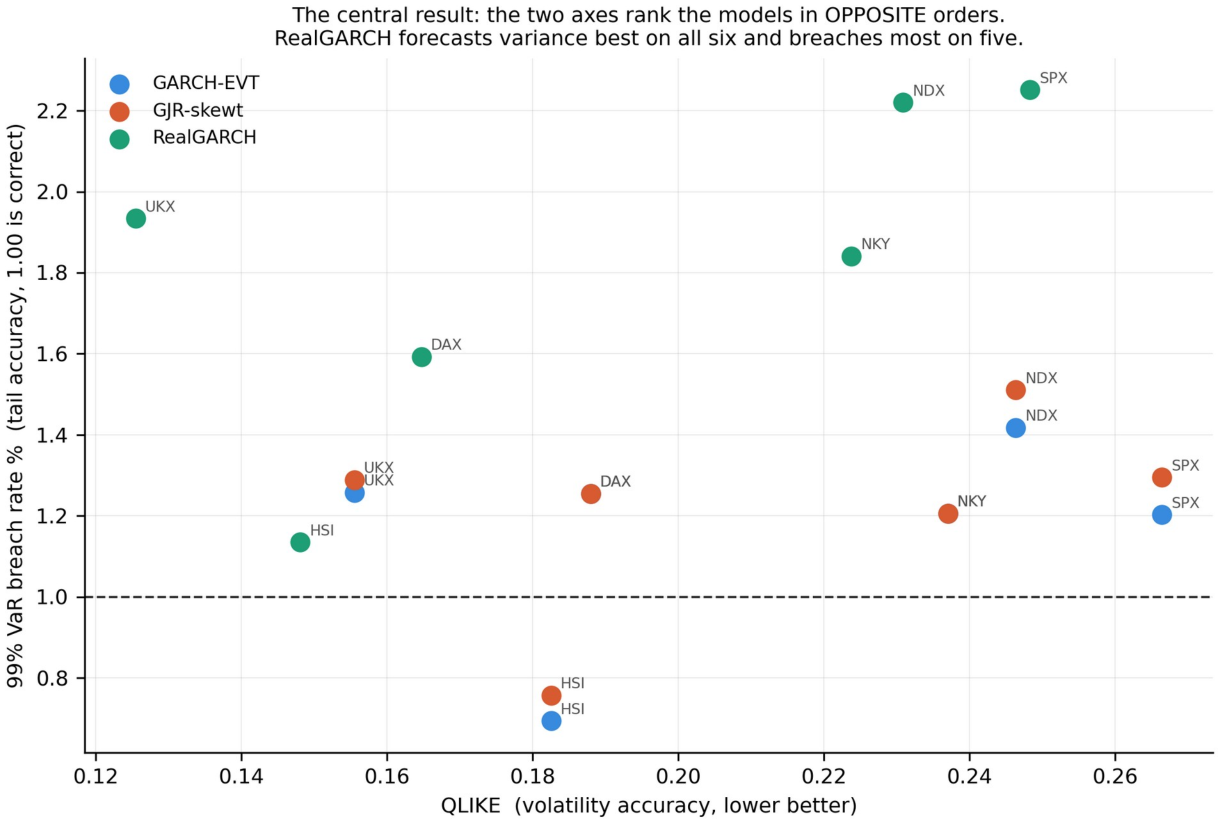
$$\text{Good } h_{(t+1)} \text{ forecast does not imply good } VaR_{(\alpha,t+1)} \text{ forecast}$$

The present result tables show Realized GARCH with lower QLIKE than the GJR benchmark across all six markets, while its 1% VaR breach rates are substantially too high in several markets. By contrast, GARCH-EVT typically has better tail coverage but shares the same variance forecast as GJR.

The core decomposition explaining this is:

$$VaR_{(\alpha,t+1)} = \mu_{(t+1)} + \sigma_{(t+1)} * q_{(\alpha,t+1)}$$

Realized information can improve the estimate of sigma without fixing a misspecified innovation tail q_alpha. This distinction is the paper's most promising contribution.



Current headline diagnostic: volatility accuracy (QLIKE) versus 99% VaR breach rate. This figure should be regenerated from the final frozen release before publication.

5. Build one authoritative final release

Before final writing, the project needs one computational source of truth. At present, the Drive contains a complete B-side result archive from one date and later corrected versions of several scripts, including the causal EVT repair. The risk is that a manuscript could accidentally combine numbers from different computational states.

Recommended release structure

```
FINAL_PAPER_RELEASE/
  code/
  forecasts/
  tables/
  figures/
  data_documentation/
  literature/
  environment/
  manuscript/
```

All final paper tables and figures should be regenerated from this single release. Older result bundles should be preserved as dated archives rather than overwritten.

6. Empirical work required before results are frozen

Priority	Action	Why it matters
P0	Consolidate latest A + B code and regenerate one complete release	Prevents version drift and mixed computational states
P0	Make Realized GARCH parameter estimation genuinely out-of-sample	Removes the main remaining parameter look-ahead concern
P0	Make Hansen-Lunde scaling causal / pre-OOS	Removes future Sample-B information from realized-volatility scaling
P1	Add Realized-GARCH-skew-t robustness	Separates variance-dynamics performance from symmetric-t tail misspecification
P1	Correct NKY post-Nov-2024 session close	Repairs the final 30 minutes of Tokyo realized-volatility measurement
P1	Add NKY missing-RV robustness	Quantifies the impact of the 2016-17 intraday-feed outage
P1	Use HAC/Newey-West inference for DM tests	Allows serial correlation in loss differentials
P1	Add EVT exceedance-dependence diagnostic	Checks an assumption not covered by marginal GPD goodness-of-fit
P2	Correct or drop the subsampled-RV claim if implementation is not truly subsampled	Avoids overstating a secondary robustness measure
P2	Verify 5-day horizon metadata and audit it	Keeps the forecast contract internally consistent
P2	Regenerate documentation and writer handoff	Ensures all narrative documents match the final release

7. Priority fix: true out-of-sample Realized GARCH

This is the most important remaining methodological issue. The GJR forecasting engine is genuinely walk-forward, but the current Realized GARCH implementation estimates a single parameter vector from the available realized-measure history and then recursively propagates the latent variance.

GJR: theta_hat_t = theta_hat(data available through t)

Current Realized GARCH: theta_hat approximately fixed using a much longer sample

Because Realized GARCH is presently the volatility-loss winner, a reviewer could reasonably argue that it benefits from information from the evaluation period when estimating its parameters.

Recommended repair

- Use expanding-window Realized GARCH parameter re-estimation.
- Annual refitting is the minimum defensible choice and may be sufficient.
- Quarterly refitting is preferable if computationally practical.
- Within each refit block, update the latent state daily using only information available at the forecast origin.

If the QLIKE advantage survives, the headline result becomes substantially stronger. If it shrinks, the corrected result should replace the current one rather than be treated as an inconvenient robustness check.

8. Priority fix: causal Hansen-Lunde scaling

The scale correction itself is economically justified: cash-session realized variance omits overnight movement, whereas the target daily return is close-to-close. The current analysis pipeline therefore rescales realized variance using a Hansen-Lunde factor.

$$c = \mathrm{sum}(r_t^2) / \mathrm{sum}(RV_t)$$

The remaining issue is timing. A single factor estimated over the full evaluation period uses future Sample-B observations when scaling early Sample-B realized variance.

Recommended alternatives

- Expanding causal scale: c_t estimated using observations strictly before the target date.
- Or estimate one fixed calibration factor on an initial pre-OOS sample and freeze it.

The expanding causal version is preferable because it preserves the spirit of the current data construction while removing future information.

9. Realized GARCH skew-t robustness

The current Realized GARCH return equation uses a symmetric Student-t innovation. Yet the baseline model-selection exercise favors skewed heavy-tailed innovations, and asymmetric return behavior is one of the project's own EDA findings.

Therefore current Realized GARCH VaR performance combines two effects:

$$\text{variance dynamics} + \text{innovation-tail specification}$$

A skew-t Realized GARCH robustness model can separate them.

Interpretation logic

- If skew-t materially improves coverage, the realized-information state estimate was useful but the symmetric tail density was too restrictive.
- If skew-t changes little, the variance-versus-tail distinction becomes even stronger.

The original Student-t Realized GARCH should remain the pre-specified main model; skew-t should be reported as a robustness extension.

10. Nikkei data-quality experiment and Tokyo session correction

2016-17 realized-measure outage

The Nikkei return series remains valid during the 2016-17 intraday-feed outage, but the realized measure is unavailable. The Realized GARCH implementation keeps those return observations and substitutes the model's own previous conditional variance inside the recursion while skipping the missing measurement term in the likelihood.

This should be turned into a transparent robustness experiment rather than buried as a footnote:

- All six markets.
- Five markets excluding NKY.
- Within NKY: observed-RV days versus recursion-imputed days.

Also correct the wording in the final limitations section: Realized GARCH tail forecasts can still be evaluated during the gap because returns exist; what is unavailable is the realized-volatility proxy needed for RV-based volatility-loss evaluation.

Tokyo trading-session change

After 5 November 2024, the Tokyo Stock Exchange closes at 15:30 rather than 15:00. Any session definition that still ends at 15:00 omits the final 30 minutes from post-change NKY realized measures. Correcting this before the final release is straightforward and removes an avoidable data-quality criticism.

11. Statistical-inference improvements

Diebold-Mariano with HAC

The current Diebold-Mariano implementation uses the variance of the one-period loss differential when h=1. However, forecast-loss differentials can remain serially correlated even for one-step-ahead forecasts.

$$\mathrm{Var}(\bar{d}) \text{ uses } \gamma_0 + 2 \sum_{l=1}^{L} w_l \gamma_l$$

Use a Newey-West/HAC long-run variance for the DM statistic. The preliminary pattern suggests that the strongest Realized-GARCH QLIKE advantages should remain significant while marginal markets remain marginal.

Why this matters for the paper

A more conservative inference procedure is preferable because it makes any surviving significance more credible and prevents the paper from relying on artificially small standard errors.

12. EVT dependence diagnostics

The marginal GPD diagnostics are already useful: threshold stability, probability-integral-transform goodness-of-fit, QQ plots, and expanding-window tail-index paths. One additional diagnostic should test whether extreme standardized residuals remain temporally clustered.

$$I_t = 1\{ -z_t > u \}$$

- Ljung-Box test on the exceedance indicator.
- Runs or first-order dependence test.
- Optional extremal-index estimate if clustering is material.

If dependence is detected, either add a declustering robustness check or state the residual clustering transparently as a limitation. This is more important than adding another exotic tail estimator.

13. Final model registry

Paper label	Specification	Role
GJR-ST	AR(1)-GJR-GARCH-skew-t	Strong parametric benchmark
GJR-EVT	Same GJR scale + POT/GPD left tail	Principal conditional EVT model
RGARCH-T	Realized GARCH + Student-t	Pre-specified realized-information model
QR-Full	13-predictor direct quantile regression	Rich-information QR
QR-Range	5-predictor sparse quantile regression	Sparse / long-history QR
RGARCH-ST	Realized GARCH + skew-t	Robustness only

After this registry is frozen, do not continue adding models unless a current specification proves invalid. Otherwise the project risks drifting into post-hoc model mining.

14. Research questions

RQ1 - Realized information and volatility

Does incorporating high-frequency realized information improve one-step-ahead conditional-variance forecasts relative to a strong asymmetric returns-only GARCH benchmark?

RQ2 - Incremental EVT value

Does POT-EVT materially improve extreme downside-risk forecasts once volatility has already been filtered through an asymmetric skewed-t GARCH model?

RQ3 - Direct conditional quantiles

Can direct quantile regression using realized, implied, range, asymmetry and macro-financial information forecast lower-tail return quantiles as accurately and reliably as distribution-based models?

RQ4 - Crises

How do model rankings change during sudden extreme market events relative to normal periods?

RQ5 - Volatility level versus onset speed

Is tail-model deterioration explained simply by the level of volatility, or does the speed of market-stress onset matter separately?

RQ6 - Forecast objective

Do models ranked highly by proper scoring rules also exhibit superior statistical and regulatory calibration?

Recommendation on hypotheses

Do not manufacture directional hypotheses after observing the results. Keep the formal structure as research questions, and discuss prior theoretical expectations from the literature without pretending they were pre-registered.

15. Final manuscript architecture

Section	Purpose
1. Introduction	Problem, two forecast targets, contribution, research questions
2. Literature Review and Research Positioning	Build the intellectual map and define the gap
3. Data and Empirical Design	Sources, markets, realized measures, sample construction, information timing
4. Econometric Models	GJR-ST, GJR-EVT, Realized GARCH, QR
5. Forecasting and Evaluation Framework	Walk-forward design, QLIKE, VaR/ES tests, DM/MCS
6. Empirical Results	Specification evidence, volatility, tails, crises, overall comparison
7. Robustness and Sensitivity	Specification, measurement, design, data-quality, tail robustness
8. Discussion and Limitations	Interpretation, what the findings mean, what they do not establish
9. Conclusion	Direct answers to the research questions

This structure is preferable to keeping separate 'Theoretical Framework' and 'Methodology' sections because it avoids repeating the same model equations twice.

16. Section-by-section writing plan

16.1 Introduction

Target approximately 1,000-1,300 words.

- Motivate volatility clustering, leverage, heavy tails, and regime instability.
- Separate conditional-scale estimation from conditional-tail estimation.
- Introduce the three modelling philosophies rather than presenting a shopping list of models.
- State the empirical contribution as a harmonized six-market comparison under common one-step-ahead evaluation.
- End with the research questions and a concise roadmap.

Core introductory contrast
GARCH-EVT = variance filtering + residual-tail extrapolation; Realized GARCH = latent volatility + high-frequency measurement; Quantile Regression = direct conditional-tail estimation.

16.2 Literature Review and Research Positioning

Subsection	Core literature stream
2.1	ARCH/GARCH foundations
2.2	Asymmetry and heavy-tailed innovations
2.3	Conditional EVT / POT
2.4	Realized volatility and microstructure
2.5	Realized GARCH and realized-information models
2.6	Conditional quantiles and CAViaR-style approaches
2.7	VaR/ES forecast evaluation
2.8	Closest comparative and hybrid evidence
2.9	Research gap and positioning

The gap should not claim that no prior study combines these ingredients. A credible formulation is that the relative ranking of these modelling philosophies is less settled when variance accuracy, tail calibration, proper scoring, and crisis-state performance are evaluated separately under one common protocol.

16.3 Data and Empirical Design

- Describe the actual free data sources, not the original proposal's paid-data plan.
- Separate daily-return history, intraday realized-measure history, and common OOS evaluation window.
- Explain per-index exchange calendars and why return-level pooling is avoided.
- Explain 5-minute realized variance, data-quality gating, half-day handling, and Hansen-Lunde scaling.
- State explicitly that returns are not winsorized, trimmed, or de-jumped because the tail is the object of study.
- Disclose that the common multi-model evaluation does not include the 2008 GFC.

16.4 Econometric Models

GJR-skew-t benchmark

$$r\_t = mu\_t + epsilon\_t, \quad epsilon\_t = sqrt(h\_t)\ z\_t$$

$$h\_t = omega + alpha\ epsilon\_(t-1)^2 + gamma\ I(epsilon\_(t-1)<0)epsilon\_(t-1)^2 + beta\ h\_(t-1)$$

Explain why skew-t is the benchmark rather than Gaussian GARCH.

GARCH-EVT

$$L\_t = -z\_t, \quad L\_t - u \mid L\_t > u \sim GPD(xi, beta)$$

State explicitly that GJR-EVT shares its conditional mean and variance with GJR-ST and differs only in the tail distribution. The final manuscript should use the corrected causal residual construction.

Realized GARCH

$$log\ h\_t = omega + beta\ log\ h\_(t-1) + gamma\ log\ x\_(t-1)$$

$$log\ x\_t = xi + phi\ log\ h\_t + tau\_1\ z\_t + tau\_2(z\_t^2 - 1) + u\_t$$

Describe the final causal parameter-refit schedule only after it has been implemented and frozen.

Quantile regression

$$Q\_tau(r\_t \mid X\_(t-1)) = X\_(t-1)'\ beta\_tau, \quad tau\ in\ \{0.01, 0.025, 0.05\}$$

Explain the forecasting lag, training-window-only standardization, separate quantile fits, and reported/repaired quantile crossings.

16.5 Forecasting and Evaluation Framework

- Common one-step-ahead convention with OriginDate < Date.
- Per-index calendars; pairwise tests use each model pair's valid-date intersection.
- Primary volatility loss: QLIKE; RMSE/MAE secondary.
- VaR: breach rate, Kupiec, Christoffersen independence and conditional coverage, DQ, pinball loss.
- ES: only where genuine ES forecasts exist.
- Statistical comparison: HAC Diebold-Mariano + Model Confidence Set.

The manuscript should clearly distinguish calibration tests from proper scoring rules.

16.6 Empirical Results

1. Specification evidence: leverage, heavy tails, skewness, residual diagnostics, and GPD threshold/tail-shape evidence.
2. Volatility forecasting: QLIKE first, with Realized GARCH versus GJR-ST.
3. Tail-risk calibration: 1% VaR first; 2.5% and 5% secondary.
4. Quantile regression behavior: sparse versus rich specification, crossings, depth-of-tail degradation.
5. Crisis performance: named crisis windows.
6. Named crises versus ex-ante volatility regimes.
7. Overall comparison: variance accuracy versus tail calibration, proper score versus coverage.

16.7 Robustness and Sensitivity

Organize robustness by threat to inference rather than as a chronological list of scripts.

Threat	Robustness checks
Specification	GJR vs EGARCH; normal vs t vs skew-t; RGARCH-T vs RGARCH-ST
Measurement	5/10/15/30-minute RV; causal scale; alternative robust measures if needed
Forecast design	Refit cadence; expanding vs fixed windows; 5-day horizon
Data quality	NKY exclusion; observed-RV vs imputed-recursion days
Tail estimation	Threshold sensitivity; exceedance dependence; largest-exceedance diagnostics
Historical reach	Pre-2013 QR-Range daily-only robustness, kept outside the main multi-model comparison

16.8 Discussion and Limitations

Use the VaR decomposition to explain why volatility and tail rankings differ. Then distinguish methodological limitations from data limitations and finite-sample limitations.

- No common three-family GFC evaluation.
- Intraday realized measures come from CFD proxies rather than the cash index itself.
- Some implied-volatility series are regional proxies.
- NKY has a long realized-measure outage.
- Extreme-tail sample sizes remain finite even with expanding EVT windows.
- Short crisis windows limit formal power.
- QR produces VaR but not a genuine variance forecast or ES.
- Model coverage differs because information sets differ.

16.9 Conclusion and Abstract

The conclusion should answer the research questions directly. The abstract should be written last, after the final rerun, and should report only frozen numbers.

17. Main figures and main tables

Recommended main figures

Figure	Purpose
1	Return-tail / leverage evidence
2	GPD threshold and QQ diagnostics

Figure	Purpose
3	Forecast versus realized volatility
4	1% VaR forecast and breach comparison
5	Crisis heatmap
6	QLIKE versus 1% breach rate - central conceptual figure
7 (optional)	QR calibration and crossing diagnostic

Recommended main tables

Table	Content
1	Markets, sources, sample periods, realized-data availability
2	Model definitions, information sets, outputs
3	Selected specification diagnostics / GPD tail parameters
4	Volatility losses with QLIKE primary
5	1% VaR breach rate and UC/CC/DQ diagnostics
6	Crisis and regime breach performance
7	Pinball / DM / MCS plus calibration summary

Full parameter tables, 2.5%/5% VaR tables, extended diagnostics, and most sensitivity figures should go to the appendix.

18. Robustness section architecture

A journal-style robustness section should answer specific objections. Each robustness subsection should begin with the threat, then the alternative design, then whether the central conclusion changes.

Specification robustness

Does the conclusion depend on GJR, Student-t, or the chosen parametric innovation density?

Measurement robustness

Does the conclusion depend on the exact realized-variance frequency or the session-to-close scaling choice?

Forecast-design robustness

Does the conclusion depend on refit cadence, window length, or horizon?

Data-quality robustness

Does the NKY outage or proxy construction drive the realized-information result?

Tail robustness

Does the EVT result depend on the 95% threshold or on residual exceedance clustering?

Historical robustness

What can be learned from earlier daily-only periods without pretending they are part of the common multi-model comparison?

19. Discussion, limitations, and claims not to make

Interpretive message

The likely final discussion should argue that conditional variance estimation and conditional extreme-tail estimation are distinct statistical tasks. The paper should resist reducing the result to a single league table.

Claims that should not appear

- The models were tested through the GFC in the common multi-model comparison.
- Quantile regression is a volatility model in this implementation.
- GARCH-EVT forecasts volatility better than GJR-ST; the two share the same variance path by construction.
- A lower crisis/normal degradation ratio automatically means greater crisis robustness.
- The model with the best ES ratio is necessarily the best-calibrated ES model.
- A Model Confidence Set retaining several models means they are equivalent on every forecasting criterion.
- The causal-EVT correction can be ignored because only one market changed breach count.
- Old handoff documents outrank regenerated final CSVs.

20. Appendix architecture

Appendix	Contents
A - Data	Session calendars, variable dictionary, missing-data treatment, scale factors
B - Model derivations	Skew-t, GPD VaR/ES, Realized-GARCH likelihood, QR objective
C - Full estimates	GARCH/EGARCH/RealGARCH parameters
D - EVT diagnostics	Threshold sweep, QQ plots, tail-index path, dependence
E - VaR/ES tables	1%, 2.5%, 5%, all backtests
F - QR diagnostics	Coefficients, crossings, predictor stability
G - Robustness	Frequency, windows, refits, horizon, NKY, RGARCH-ST

Appendix	Contents
H - Reproducibility	Software versions, run order, forecast schema, audit

21. Claim ledger and source-of-truth hierarchy

Claim ledger

Before narrative drafting, create a compact evidence ledger so every manuscript sentence containing a substantive numerical claim maps to a final table and, where relevant, a figure and literature citation.

Claim ID	Example claim	Primary source	Status
C01	Realized GARCH has lower QLIKE across the six markets	Final 47a volatility-loss table	Pending fair RGARCH rerun
C02	EVT tail shape is positive across markets	Final threshold / EVT diagnostic tables	Pending final regeneration
C03	QR-Full crosses quantiles more often than QR-Range	Final QR summary	Likely stable
C04	All models under-cover sharply during COVID	Final crisis table	Pending final regeneration
C05	Proper-score rankings differ from coverage rankings	Final DM/MCS + VaR tables	Pending HAC update

Source-of-truth hierarchy

Final regenerated CSV > final figure > final audit > writer handoff > old documentation > original proposal

This hierarchy is essential because the proposal describes intended methods and data sources, while the current code describes what was actually implemented.

22. Team execution sequence

8. Create FINAL_PAPER_RELEASE and unify the codebase.
9. Correct NKY session definition and any known secondary realized-measure implementation issue.
10. Rebuild affected realized data.
11. Implement causal / expanding Hansen-Lunde scaling.
12. Implement annual or quarterly walk-forward Realized GARCH.
13. Add Realized-GARCH-skew-t robustness.
14. Add HAC DM inference and EVT exceedance-dependence diagnostics.
15. Fix horizon metadata and extend the reproducibility audit to check it.
16. Rerun all forecasts, evaluations, tables, and figures.
17. Run automated audit plus a small manual formula audit against published methods.

18. Freeze the final release and record the release date / commit / checksum.
19. Update the claim ledger.
20. Freeze the literature corpus and evidence matrix.
21. Write Data, Models, and Evaluation sections first.
22. Write Results directly from the frozen claim ledger.
23. Write Robustness, Discussion, and Limitations.
24. Write the Literature Review synthesis.
25. Write the Introduction.
26. Write the Conclusion.
27. Write the Abstract last.
28. Perform a final citation, numerical, and reproducibility audit.

23. Expected final scientific contribution

If the major current patterns survive the fair reruns, the paper will support a richer conclusion than 'one model wins'. It will show that different notions of forecast quality can point in different directions.

$$\text{Volatility accuracy} \;\; != \;\; \text{tail calibration} \;\; != \;\; \text{proper-score ranking} \;\; != \;\; \text{crisis resilience}$$

The likely conceptual contributions are:

- High-frequency realized measures can materially improve conditional-variance tracking without guaranteeing correct extreme-tail quantiles.
- A strong skewed-t GARCH benchmark may absorb much of the incremental VaR benefit often attributed to EVT.
- High-dimensional quantile regression can contain more information yet generalize worse in the deepest tail than a sparse specification.
- Abrupt crisis onset can be more damaging to calibration than a sustained high-volatility regime that the recursion has time to absorb.
- Calibration tests and proper scoring rules answer different questions and need not rank models identically.

The paper's strongest potential message
There is no single notion of the 'best' tail-risk model. A forecasting system should be judged against the object it is designed to estimate and the decision criterion for which it will be used.

24. Immediate action checklist

Step	Action	Status
1	Freeze current Drive state and preserve dated archive	Not started
2	Create unified FINAL_PAPER_RELEASE tree	Not started

Step	Action	Status
3	Correct NKY post-Nov-2024 trading session	Required
4	Implement causal Hansen-Lunde scaling	Required
5	Implement walk-forward Realized GARCH	Required
6	Add Realized-GARCH-skew-t robustness	Recommended
7	Replace DM variance with HAC/Newey-West	Required
8	Add EVT exceedance-dependence diagnostic	Recommended
9	Run NKY missing-RV robustness	Required
10	Regenerate forecasts / tables / figures / audit	Required
11	Freeze final result release	Required
12	Start manuscript drafting from Sections 3-5	After freeze

Final recommendation

Do not polish the current Results narrative or Abstract yet. First make the remaining methodological corrections, regenerate one authoritative release, and freeze the empirical evidence. Then write the paper around the distinction between conditional-scale accuracy and extreme-tail calibration, with every result traceable to a final CSV and every interpretation traceable to the literature.

Evidence base reviewed for this plan

This plan was prepared from the current project documentation, model code, forecast contract, evaluation scripts, result tables, and figure set in the project's Google Drive. The most important implementation documents include the analysis-dataset builder, Realized GARCH implementation, GARCH-EVT implementation, quantile-regression implementation, evaluation script, crisis/regime analysis, and model-comparison script.

- 15_build_analysis_dataset.py
- 28_realized_garch.py
- 42_garch_evt.py
- 44_qr.py
- 47_evaluation.py
- 48_crisis_regime.py
- 49_model_comparison.py
- 50_reproducibility_audit.py